

GEOMETRIC STRUCTURES ON GRAPHS: A HOLONOMY-BASED DISCRETIZATION OF CURVATURE

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ABSTRACT. We propose a holonomy-based framework for discretizing curvature on graphs equipped with local symmetric positive-definite metrics. Each vertex carries a fibre metric g_i , and each directed edge carries a reversible metric-compatible transport F_{ij} . The ordered product around an oriented triangular loop $\mathcal{C}$ gives a holonomy $H_{\mathcal{C}}$, whose normalized logarithm $\Omega_{\mathcal{C}} = -s_{\mathcal{C}}^{-1} \text{Log}(H_{\mathcal{C}})$ is used as a finite-loop curvature observation. Thus the construction discretizes the geometric principle that infinitesimal holonomy is controlled by curvature, rather than treating holonomy as a heuristic feature. Since $\Omega_{\mathcal{C}}$ lies in the g_i -orthogonal Lie algebra, it is not itself a velocity of an SPD metric. We therefore introduce two aggregation mechanisms: a commutator with a symmetric response matrix, producing symmetric Ricci-type metric responses, and an incidence-aware covariant divergence of curvature-induced edge fluxes, reflecting the relation between trace and covariant divergence. The resulting responses are locally orthogonal-gauge equivariant and can drive exponential updates that preserve positive definiteness. We also give a reversible metric-compatible parametrization of edge transports, allowing orthogonal edge factors, loop scales, weights, and response matrices to be learned while respecting the graph geometry. Known-geometry calibrations on the unit sphere test the holonomy–curvature relation, curvature preservation under nontrivial local metric representations, and the empirical recovery of edge transports from local observations.

1. INTRODUCTION

Curvature has provided an important geometric language for network analysis and graph learning. Ollivier–Ricci curvature compares neighbourhood probability measures by optimal transport [9], while Forman–Ricci curvature arises from a Bochner–Weitzenböck type principle on weighted cell complexes [4]. Under suitable sampling and scaling assumptions, rescaled Ollivier–Ricci curvature on random geometric graphs can also converge to the Ricci curvature of an underlying Riemannian manifold [12]. These notions and their variants have been used for community detection and graph Ricci flow [8], over-squashing and curvature-based rewiring [11], and curvature-weighted message passing in graph learning models [7].

In many graph-learning uses, however, curvature enters the model after being compressed into a scalar descriptor on edges or vertices, where it modulates neighborhood weights, rewiring rules, or information-propagation strength. This effectively exploits local structural information revealed by curvature, but it does not directly discretize the tensor algebra or geometric construction mechanism of curvature itself. This leads to the question addressed in this paper: instead of starting from a prescribed scalar curvature, can graph interactions themselves be organized as discrete geometric data with local metrics and edgewise parallel transports, so that curvature is obtained from their failure to compose trivially around local closed loops?

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We study this question through the relation between local metric structures, edgewise parallel transport, and holonomy on graphs. Each vertex i is assigned a fibre $E_i \cong \mathbb{R}^n$ with a positive-definite metric $g_i \in \text{SPD}_n$, and a linear map on a directed edge $j \rightarrow i$,

$$F_{ij} : E_j \rightarrow E_i,$$

is interpreted as a discrete parallel transport from the local fibre at j to the local fibre at i . Metric compatibility requires

$$F_{ij}^\top g_i F_{ij} = g_j, \quad F_{ji} = F_{ij}^{-1}.$$

This structure can be viewed as a finite gluing of locally flat patches: vertices aggregate local coordinate neighborhoods, edges encode finite identifications between neighboring patches, and curvature is detected by the nontrivial holonomy produced when these identifications are composed around closed loops. More abstractly, it equips a graph with geometrically constrained gauge connection data.

The construction is motivated by a classical fact in differential geometry: for a small loop shrinking to a point, the holonomy of parallel transport is controlled at second order by the curvature of the connection. Thus, if $\mathcal{C} = (i, j, k, i)$ is a selected oriented triangular loop, its holonomy

$$H_{\mathcal{C}} = F_{ik} F_{kj} F_{ji}$$

is an automorphism of the base fibre. Whenever the principal logarithm is defined, the normalized quantity

$$\Omega_{\mathcal{C}} = -\frac{1}{s_{\mathcal{C}}} \text{Log}(H_{\mathcal{C}})$$

gives a curvature observation at a finite loop scale. Here $s_{\mathcal{C}}$ is an effective loop scale compatible with the graph geometry: when embedding or area information is available, it may be induced by a local area; on an abstract graph, it is an explicitly specified normalization structure. The purpose is therefore not to claim a unique curvature for a purely combinatorial graph, but to discretize the continuous holonomy–curvature mechanism into computable and composable graph-level objects.

The curvature observation $\Omega_{\mathcal{C}}$ is g_i -skew-adjoint, whereas a velocity of a positive-definite metric must be symmetric. To connect these two types of objects, we introduce a symmetric response matrix $B_{\mathcal{C},i}$ and define a Ricci-type response through the commutator

$$\text{Ric}_g^D(i) = \sum_{\mathcal{C} \in \mathcal{C}_i} \alpha_{\mathcal{C}} [B_{\mathcal{C},i}, g_i \Omega_{\mathcal{C}}].$$

This commutator does not redefine curvature. Rather, it describes a noncommutative coupling between the local curvature channel and the metric-response channel, thereby producing a direction that can drive an update of the positive-definite metric.

In addition to directly aggregating loop curvature at a vertex, we also preserve the incidence structure between oriented faces and edges. More precisely, loop curvature first induces an antisymmetric edge flux, which is then aggregated at vertices by a covariant graph divergence. This mechanism corresponds to the continuous relation

$$\text{div } V = \text{tr}_g(\nabla V),$$

where local geometric information is collected through a covariant derivative followed by a trace. It is related to, but distinct from, the contraction of the curvature tensor that produces the Ricci tensor: the former gives a curvature aggregation that retains face–edge incidence information, while the latter requires additional directional and quadrature structures. We keep these two mechanisms separate, so that direct aggregation and divergence aggregation can support different graph-geometric interactions.

The resulting structure can be naturally embedded into graph-learning layers. At each layer, orthogonal factors O_{ij} generate strictly metric-compatible edge transports by

$$F_{ij} = g_i^{-1/2} O_{ij} g_j^{1/2}.$$

Loop holonomy then produces curvature observations; Ricci-type responses evolve the family $\{g_i\}$ through an exponential update that preserves positive definiteness; and semantic features may be propagated by message-passing operators conditioned on curvature invariants or other geometric statistics. In this way, metric evolution and semantic propagation are separated into two coupled but distinct stages: the former constructs and updates local geometry, while the latter uses that geometry to modulate task-dependent information exchange.

The main contributions are as follows.

- We propose a graph geometric structure in which vertices carry local SPD metrics and edges carry reversible metric-compatible discrete parallel transports, together with an orthogonal-factor parametrization and local orthogonal gauge symmetry.
- Starting from the continuous small-loop holonomy expansion, we construct normalized logarithmic holonomy as a graph curvature observation, and give both direct loop aggregation and a covariant divergence aggregation based on curvature-induced edge fluxes.
- We use a commutator construction to resolve the mismatch between the skew-adjointness of curvature observations and the symmetry required of metric velocities, obtaining symmetric gauge-equivariant Ricci-type responses and an SPD-preserving discrete update.
- We provide a structure-preserving learning realization, and use known-geometry calibrations on the unit sphere to test the holonomy–curvature relation, curvature preservation under metric-compatible representations, and the empirical trend of recovering edge transports from local observations.

The rest of the paper is organized as follows. Section 2 defines local metric fibres, edge transports, and oriented loops on graphs. Section 3 constructs holonomy-based curvature observations and gives both direct and divergence-type aggregations. Section 4 defines Ricci-type responses and SPD-preserving metric evolution. Section 5 discusses a structure-preserving learning realization. Section 6 gives basic structural checks and special cases. Section 7 reports known-geometry calibrations and synthetic consistency experiments.

2. GRAPH GEOMETRIC STRUCTURES

2.1. Local metric fibres and edge transport. Let $G = (V, E)$ be a finite undirected graph. We write $j \sim i$ when $\{i, j\} \in E$, and replace each undirected edge by both directed edges. At every node i , let $E_i \cong \mathbb{R}^n$ be a fibre equipped with the inner product

$$\langle u, v \rangle_{g_i} = u^\top g_i v, \quad g_i \in \text{SPD}_n.$$

The matrices g_i may be interpreted as node features, as local metric data, or as learned latent metrics. The use of an SPD feature space is consistent with the affine-invariant geometry of positive-definite matrices [1].

To motivate a geometric realization, one may regard the graph as encoding a *piecewise-flat* underlying space: a node i represents a locally flat patch U_i equipped with a chosen parallel frame, and in that frame the metric is represented by a constant matrix $g_i \in \text{SPD}_n$. An edge records the adjacency of two such patches and carries a finite identification of their frames. The corresponding curvature is not asserted to occur inside an individual flat patch; it is detected by the failure of these identifications to compose trivially around a loop encircling interfaces or hinges.

Under this interpretation, F_{ij} is a discrete parallel transport from the fibre over j to the fibre over i . It is not, in general, the coordinate transition Jacobian $\varphi_i \varphi_j^{-1}$: the latter satisfies an atlas cocycle relation on triple overlaps, whereas the former is the connection data whose loop product records curvature. A metric-compatible family of such finite transports, supplemented when appropriate by a discrete torsion-free condition, is therefore Levi–Civita-inspired in this piecewise-flat realization. More generally, the properties of a connection in our framework are encoded through the parallel transports it induces along graph edges.

Definition 2.1 (Metric graph connection). *A metric graph connection on $(G, \{g_i\})$ is a family of invertible edge maps*

$$F_{ij} : E_j \longrightarrow E_i, \quad j \sim i,$$

such that

$$(1) \quad F_{ij}^\top g_i F_{ij} = g_j, \quad F_{ji} = F_{ij}^{-1}.$$

The first condition is the discrete counterpart of metric compatibility: the transport preserves fibre inner products. It is a modelling constraint, not a consequence of an arbitrary choice of edge maps.

Proposition 2.1 (Orthogonal-equivariant). *For each directed edge $j \rightarrow i$, the metric compatibility condition in (1) holds if and only if*

$$(2) \quad F_{ij} = g_i^{-1/2} O_{ij} g_j^{1/2} \quad \text{for some } O_{ij} \in O(n).$$

Moreover, edge reversal is enforced by choosing $O_{ji} = O_{ij}^\top$.

Proof. Set $O_{ij} = g_i^{1/2} F_{ij} g_j^{-1/2}$. Equation (1) gives $O_{ij}^\top O_{ij} = I$, proving $O_{ij} \in O(n)$ and (2). Conversely, direct substitution of (2) gives $F_{ij}^\top g_i F_{ij} = g_j$. The inverse relation follows from $O_{ji} = O_{ij}^\top$. $\square$

In the piecewise-flat realization above, F_{ij} may be viewed as a finite parallel transport between adjacent locally flat patches. On an abstract graph this interpretation is additional geometric structure rather than automatic data; Definition 2.1 nevertheless remains meaningful as a metric-compatible discrete gauge connection.

For completeness, we retain an optional discrete analogue of torsion freeness. In Cartan's moving-frame notation, the torsion of a connection A is

$$\Theta(A) := de + \omega_A \wedge e,$$

where e is the frame field and ω_A is the connection one-form. Thus the torsion-free condition is $\Theta(A) = 0$. Suppose now that each directed edge $i \rightarrow j$ is assigned a displacement vector $v_{ij} \in E_i$. For a triangular loop based at i , the three terms below all belong to E_i .

Definition 2.2. *If the graph assigns each directed edge $i \rightarrow j$ a displacement vector $v_{ij} \in E_i$, the first structure equation motivates the weak torsion-free condition for the parallel transports $\{F_{ij}\}$ that, for every triangular loop*

$$(i, j, k) := \{i \rightarrow j \rightarrow k \rightarrow i \mid i, j, k \in V\},$$

$$(3) \quad v_{ij} + F_{ij} v_{jk} + F_{ik} v_{ki} = 0.$$

For abstract graphs without a spatial embedding, the displacement vectors and condition (3) are additional modelling data and should not be imposed by default. We therefore retain metric compatibility as the standing requirement. Dropping the optional weak torsion-free condition gives a general metric-compatible connection, while learning the edge factors O_{ij} may model geometric effects not represented by the abstract graph. In the piecewise-flat interpretation, imposing both compatibility and the weak closure condition is the sense in which the construction most closely resembles a discrete Levi-Civita connection.

2.2. Loops, scales, and gauge changes. We use selected triangular faces as minimal graph loops. Let

$$\mathcal{T} := \{\{i, j, k\} \subseteq V : \{i, j\}, \{j, k\}, \{k, i\} \in E\}$$

be the collection of underlying triangular faces of G . For every $\tau = \{i, j, k\} \in \mathcal{T}$, choose exactly one of its two cyclic orientations. The resulting oriented face collection is denoted by $\mathcal{F} \subseteq V^3$. Thus, for each $\tau \in \mathcal{T}$, exactly one of (i, j, k) and (i, k, j) , up to cyclic permutation, belongs to $\mathcal{F}$. The ordered triple $\sigma = (i, j, k) \in \mathcal{F}$ represents the oriented closed walk $\mathcal{C} = (i, j, k, i)$. If $i \in \sigma$, write $\sigma_i = (i, j, k)$ for the cyclic re-rooting of σ whose first vertex is i , and write

$$\mathcal{C}_{\sigma, i} := (i, j, k, i)$$

for the corresponding rooted closed loop. Thus σ emphasizes the oriented face, while $\mathcal{C}_{\sigma, i}$ emphasizes the associated loop and its base fibre.

For $\sigma_i = (i, j, k)$, define the holonomy of σ based at i by

$$(4) \quad H_{\sigma_i} = H_{\mathcal{C}_{\sigma,i}} : E_i \longrightarrow E_i, \quad H_{\sigma_i} := F_{ik}F_{kj}F_{ji}.$$

Metric compatibility implies $H_{\sigma_i}^\top g_i H_{\sigma_i} = g_i$. Thus loop holonomy lies in the g_i -orthogonal group. We use the principal matrix logarithm only under the standard admissibility condition that H_{σ_i} has no eigenvalue on the closed negative real axis. This condition excludes the branch ambiguity at -1 and guarantees a real logarithm compatible with the chosen local branch.

The relation between holonomy and baseline geometry is stressed in the following lemma.

Lemma 1. *Let $\pi : (E, \nabla) \rightarrow M$ be a smooth vector bundle with connection, let $p \in M$, and let $X, Y \in T_p M$. Choose a smooth local surface Φ with*

$$\Phi(0, 0) = p, \quad \partial_s \Phi(0, 0) = X, \quad \partial_t \Phi(0, 0) = Y.$$

For $\varepsilon > 0$, let γ_ε be the positively oriented boundary of $\Phi([0, \varepsilon]^2)$, based at p , and denote by $P_{\gamma_\varepsilon} : E_p \rightarrow E_p$ the parallel transport around γ_ε . Fix the curvature convention

$$R^\nabla(X, Y) := \nabla_X \nabla_Y - \nabla_Y \nabla_X - \nabla_{[X, Y]}.$$

With the orientation and parallel-transport convention chosen so that the displayed sign holds, one has

$$P_{\gamma_\varepsilon} = \text{Id} - \varepsilon^2 R_p^\nabla(X, Y) + O(\varepsilon^3).$$

Consequently, for sufficiently small ε ,

$$\text{Log}(P_{\gamma_\varepsilon}) = -\varepsilon^2 R_p^\nabla(X, Y) + O(\varepsilon^3).$$

Proof. This is the classical local holonomy formula; see, for example, [6, Chapter II]. The logarithmic expansion follows from $\text{Log}(I + A) = A + O(\|A\|^2)$ as $A \rightarrow 0$. $\square$

Lemma 1 is the continuous mechanism underlying our construction: curvature is recovered from the normalized logarithmic holonomy of a shrinking loop. Selected graph loops are finite proxies for such local loops. Accordingly, for a selected loop $\mathcal{C}$ with holonomy $H_{\mathcal{C}}$, we represent the curvature tensor acting on the in-and-out direction pair of loop $\mathcal{C}$ by

$$\Omega_{\mathcal{C}} := -\frac{1}{s_{\mathcal{C}}} \text{Log}(H_{\mathcal{C}}),$$

where $s_{\mathcal{C}} > 0$ is an effective curvature scale determined by the chosen graph geometry. When a local embedding or a controlled cover is available, $s_{\mathcal{C}}$ may be chosen from the corresponding positive (unsigned) area scale; otherwise it is specified through the loop normalization.

Definition 2.3 (Admissible loop curvature). *For an admissible oriented face $\sigma = (i, j, k) \in \mathcal{F}$, its re-rooting $\sigma_i = (i, j, k)$, and the associated loop $\mathcal{C}_{\sigma,i} = (i, j, k, i)$, set*

$$\Xi_{\sigma,i} := \text{Log}(H_{\sigma,i}), \quad \Xi_{\mathcal{C}_{\sigma,i}} := \text{Log}(H_{\mathcal{C}_{\sigma,i}}) = \Xi_{\sigma,i}.$$

Let $s_\sigma = s_{\mathcal{C}_{\sigma,i}} > 0$ be an effective loop scale. The normalized loop-curvature observation based at i is

$$\Omega_{\sigma,i} := -\frac{1}{s_\sigma} \Xi_{\sigma,i}, \quad \Omega_{\mathcal{C}_{\sigma,i}} := -\frac{1}{s_{\mathcal{C}_{\sigma,i}}} \Xi_{\mathcal{C}_{\sigma,i}} = \Omega_{\sigma,i}.$$

Because $H_{\mathcal{C}}$ preserves g_i , its logarithm satisfies

$$(5) \quad \Xi_{\mathcal{C}}^\top g_i + g_i \Xi_{\mathcal{C}} = 0, \quad \Omega_{\mathcal{C}}^\top g_i + g_i \Omega_{\mathcal{C}} = 0.$$

Thus $\Xi_{\mathcal{C}}$ and $\Omega_{\mathcal{C}}$ are both g_i -skew-adjoint; equivalently, $g_i \Xi_{\mathcal{C}}$ and $g_i \Omega_{\mathcal{C}}$ are skew-symmetric.

On a graph, $s_{\mathcal{C}}$ is an explicit modelling choice. If displacement vectors or an embedding provide an area $A(\mathcal{C})$, one possible effective scale is

$$(6) \quad s_{\mathcal{C}} = (2A(\mathcal{C}) + \varepsilon_A) \exp(\gamma r_{\mathcal{C}}^2), \quad \gamma, \varepsilon_A > 0,$$

where $r_{\mathcal{C}}$ measures loop size. Then the factor $s_{\mathcal{C}}^{-1}$ gives a decaying area-normalized weight. If no geometric embedding is given, the normalization should be understood as an effective graph-scale choice rather than as a literal area reciprocal.

The construction has a natural local orthogonal gauge symmetry: for $Q_i \in O(n)$ at every node, change the fibre coordinates by

$$(7) \quad g'_i = Q_i g_i Q_i^\top, \quad F'_{ij} = Q_i F_{ij} Q_j^\top.$$

Then $H'_C = Q_i H_C Q_i^\top$ and, on the selected logarithm branch, $\Omega'_C = Q_i \Omega_C Q_i^\top$. This equivariance will be inherited by the metric responses below. We restrict to orthogonal gauge changes because the commutator construction is expressed using matrix products after lowering an index; general linear coordinate covariance would require a more carefully typed tensor formulation.

3. HOLONOMY-BASED CURVATURE OBSERVATIONS

3.1. From loop holonomy to a curvature signal. For each vertex i , let

$$\mathcal{F}_i := \{\sigma \in \mathcal{F} : i \in \sigma\}, \quad \mathcal{C}_i := \{\mathcal{C}_{\sigma,i} : \sigma \in \mathcal{F}_i\}.$$

The direct holonomy aggregation is

$$(8) \quad K_i := \sum_{\sigma \in \mathcal{F}_i} \alpha_{\sigma,i} \Omega_{\sigma,i}, \quad \alpha_{\sigma,i} =: \alpha_{\mathcal{C}_{\sigma,i}} \geq 0.$$

Each selected underlying triangular face occurs exactly once in $\mathcal{F}$. Its orientation is encoded in the oriented loop defining $\Omega_{\sigma,i}$, while $\alpha_{\sigma,i}$ is an unsigned scale or confidence weight. Since each summand is g_i -skew-adjoint, K_i is also g_i -skew-adjoint. In a smooth setting, the curvature tensor is a two-form with values in endomorphisms, and the Ricci tensor arises only after a specified contraction. Equation (8) does not recover this canonical contraction on an arbitrary graph: it is an aggregated loop-curvature signal. The distinction is central to the interpretation of our method.

There are two ways to give α_C a geometric meaning. With an embedded graph and displacement vectors, a loop area and a local frame can motivate the scale in (6). Recall that under a fixed global trivialization, the node metrics admit a common matrix representation, and one may use the affine-invariant distance

$$(9) \quad \text{dSPD}(g_i, g_j) = \|\log(g_i^{-1/2} g_j g_i^{-1/2})\|_F$$

to form an effective loop size, for example

$$r_{ijk}^2 = \text{dSPD}(g_i, g_j)^2 + \text{dSPD}(g_j, g_k)^2 + \text{dSPD}(g_k, g_i)^2.$$

This option is useful when the node matrices share a reference basis. Under independent local gauges, however, the direct comparison in (9) is not an invariant quantity. It must therefore not be presented as an intrinsic gauge-invariant edge scalar without an additional registration assumption.

Proposition 3.1 (Frame-based Ricci contraction calibration). *Suppose that the graph data near a node i are obtained from a local sampling of a Riemannian manifold (M, g) , and identify E_i with $T_{p_i}M$. Let $\mathcal{N}_i^{\text{tr}} \subseteq \mathcal{N}(i)$ denote the set of neighbours selected for the discrete trace approximation, and $\{(v_{ik}, \rho_{ik})\}_{k \in \mathcal{N}_i^{\text{tr}}}$ be a weighted family in $T_{p_i}M$ satisfying the resolution of identity*

$$(10) \quad \sum_{k \in \mathcal{N}_i^{\text{tr}}} \rho_{ik} v_{ik} \otimes v_{ik}^\flat = \text{Id}_{T_{p_i}M}, \quad v_{ik}^\flat(Z) := g_i(v_{ik}, Z).$$

For each sampled direction v_{ij} and each $k \in \mathcal{N}_i^{\text{tr}}$, let $\mathcal{C}_{ikj}$ be a selected oriented local loop that admits a smooth realization as the boundary of a shrinking surface to which Lemma 1 applies. Assume that its scale is calibrated by $s_{\mathcal{C}_{ikj}} = \varepsilon_{ikj}^2 + O(\varepsilon_{ikj}^3)$. Then the lemma and the normalization in Definition 2.3 give

$$(11) \quad \Omega_{\mathcal{C}_{ikj}} = R_{p_i}(v_{ik}, v_{ij}) + E_{ikj}, \quad E_{ikj} = O(\varepsilon_{ikj}).$$

where R is the curvature of the underlying connection and E_{ikj} is the loop-curvature approximation error. Define

$$(12) \quad \widehat{\text{Ric}}_i(v_{ij}, Y) := \sum_{k \in \mathcal{N}_i^{\text{tr}}} \rho_{ik} g_i(\Omega_{\mathcal{C}_{ikj}} Y, v_{ik}).$$

Then

$$(13) \quad \widehat{\text{Ric}}_i(v_{ij}, Y) = \text{Ric}_{p_i}(v_{ij}, Y) + \sum_{k \in \mathcal{N}_i^{\text{tr}}} \rho_{ik} g_i(E_{ikj}Y, v_{ik}).$$

Proof. Let $A_Y : Z \mapsto R_{p_i}(Z, v_{ij})Y$. By (10),

$$\text{tr}(A_Y) = \sum_{k \in \mathcal{N}_i^{\text{tr}}} \rho_{ik} g_i(A_Y v_{ik}, v_{ik}).$$

Hence

$$\text{Ric}_{p_i}(v_{ij}, Y) = \sum_{k \in \mathcal{N}_i^{\text{tr}}} \rho_{ik} g_i(R_{p_i}(v_{ik}, v_{ij})Y, v_{ik}).$$

Substituting (11) yields (13). $\square$

Remark. The weights ρ_{ik} in (10) are trace-quadrature weights and should not be conflated with the edge interaction weights w_{ik} or the loop aggregation weights α_C . The proposition gives a direction-resolved discrete proxy for the Ricci contraction. By contrast, the operator K_i in (8) is a direction-compressed loop-curvature signal; the commutator construction in Definition 4.1 converts such a signal into a symmetric metric response.

3.2. An incidence-based covariant divergence. The direct aggregate (8) retains loop curvature at a vertex but discards how individual loops are incident on the neighbouring edges. We now organize the same local observations through a covariant edge-flux construction. The geometric motivation is related to, but distinct from, the Ricci contraction discussed above. On a Riemannian manifold, divergence is itself a trace operation:

$$\text{div } V = \text{tr}_g(\nabla V)$$

for a vector field V , whereas the Ricci tensor is the trace of the different endomorphism $Z \mapsto R(Z, X)Y$. Thus both operations contract local geometric information, but they contract different objects. The construction below discretizes the former mechanism: oriented faces create curvature-induced edge fluxes, which are then collected at a vertex by a covariant divergence. It is not presented as a second definition of the Ricci contraction.

We now use the oriented face collection $\mathcal{F}$ fixed above; each selected underlying triangular face occurs exactly once. If $\sigma \in \mathcal{F}$ is incident on a vertex i , let $\Omega_{\sigma,i} \in \text{End}(E_i)$ denote its normalized logarithmic holonomy based at i , obtained by cyclically re-rooting the oriented loop at i . For example, if $\sigma_i = (i, j, k)$, so that $\mathcal{C}_{\sigma,i} = (i, j, k, i)$, then

$$(14) \quad \Omega_{\sigma,i} = -\frac{1}{s_\sigma} \text{Log}(F_{ik}F_{kj}F_{ji}), \quad \Omega_{\sigma,j} = F_{ji}\Omega_{\sigma,i}F_{ij}.$$

The second identity follows from $F_{ji} = F_{ij}^{-1}$ and the similarity equivariance of the selected logarithm branch. In particular, each $\Omega_{\sigma,i}$ is g_i -skew-adjoint.

For an oriented edge $i \rightarrow j$ contained in σ , let $\epsilon_{ij}(\sigma) = 1$ when the boundary orientation of σ traverses $i \rightarrow j$, and let $\epsilon_{ij}(\sigma) = -1$ otherwise. Thus $\epsilon_{ji}(\sigma) = -\epsilon_{ij}(\sigma)$. Write $\mathcal{F}_{ij}$ for the faces containing the underlying edge $\{i, j\}$, and define the curvature edge cochain based at i by

$$(15) \quad A_{ij} := \sum_{\sigma \in \mathcal{F}_{ij}} \epsilon_{ij}(\sigma) \Omega_{\sigma,i} \in \text{End}(E_i).$$

Although the subscript ij records the graph direction $i \rightarrow j$, the coefficient A_{ij} acts on the source fibre E_i . Acting on a vector $s_i \in E_i$, the quantity $A_{ij}s_i$ is the corresponding curvature-induced outgoing edge displacement. The edge cochain A , rather than an evaluation $A_{ij}s_i$ at an arbitrary section, is the object that has the required covariant antisymmetry.

Proposition 3.2 (Covariant flux identities). *The curvature edge cochain (15) satisfies*

$$(16) \quad A_{ji} = -F_{ji}A_{ij}F_{ij}.$$

Let $w_{ij} = w_{ji} \geq 0$ be edge conductances and let $\mu_i > 0$ be vertex masses. The covariant graph divergence of A is the endomorphism

$$(17) \quad K_i^{\text{div}} := (\text{div}_{w,\mu}^F A)_i := \frac{1}{\mu_i} \sum_{j \sim i} w_{ij} A_{ij} \in \text{End}(E_i).$$

It is g_i -skew-adjoint and is equivariant under the local orthogonal gauge action (7). More precisely,

$$(18) \quad (K_i^{\text{div}})' = Q_i K_i^{\text{div}} Q_i^\top.$$

Moreover, for an endomorphism-valued vertex cochain $B = \{B_i \in \text{End}(E_i)\}_{i \in V}$, define its covariant edge increment by

$$(19) \quad (d^F B)_{ij} := F_{ij} B_j F_{ji} - B_i \in \text{End}(E_i).$$

Endow $\text{End}(E_i)$ with the g_i -Hilbert–Schmidt pairing

$$(20) \quad \langle S, T \rangle_{g_i} := \text{tr}(S^{\dagger g_i} T), \quad S^{\dagger g_i} := g_i^{-1} S^\top g_i.$$

Then, for any choice of one orientation of each undirected edge,

$$(21) \quad \sum_{i \in V} \mu_i \langle B_i, (\text{div}_{w,\mu}^F A)_i \rangle_{g_i} = - \sum_{\{i,j\} \in E} w_{ij} \langle (d^F B)_{ij}, A_{ij} \rangle_{g_i}.$$

Proof. The rooted holonomies of a fixed oriented face are related by transport as in (14). Together with $\epsilon_{ji}(\sigma) = -\epsilon_{ij}(\sigma)$, this gives

$$A_{ji} = \sum_{\sigma \in \mathcal{F}_{ij}} \epsilon_{ji}(\sigma) \Omega_{\sigma,j} = -F_{ji} \left(\sum_{\sigma \in \mathcal{F}_{ij}} \epsilon_{ij}(\sigma) \Omega_{\sigma,i} \right) F_{ij},$$

which proves (16). Since each summand in (15) is g_i -skew-adjoint, the same is true of A_{ij} and of the weighted sum (17). Gauge equivariance follows because every factor in (15) based at i transforms by conjugation with Q_i ; the scalar weights w_{ij} and μ_i are gauge invariant.

Metric compatibility makes $F_{ji} : (E_i, g_i) \rightarrow (E_j, g_j)$ an isometry. Consequently, conjugation by F_{ji} preserves the pairings (20). Using (16) and grouping the left-hand side of (21) over undirected edges yields

$$\begin{aligned} \sum_{i \in V} \mu_i \langle B_i, (\text{div}_{w,\mu}^F A)_i \rangle_{g_i} &= \sum_{\{i,j\} \in E} w_{ij} (\langle B_i, A_{ij} \rangle_{g_i} + \langle B_j, A_{ji} \rangle_{g_j}) \\ &= \sum_{\{i,j\} \in E} w_{ij} \langle B_i - F_{ij} B_j F_{ji}, A_{ij} \rangle_{g_i}, \end{aligned}$$

which is (21) by (19). $\square$

Proposition 3.2 is the discrete analogue of the divergence mechanism used here: face curvature is first assigned to its oriented boundary edges, and the signed outgoing fluxes are then summed at a vertex. Equation (21) identifies this operation as the negative adjoint of a covariant edge increment, exactly as divergence is the negative adjoint of a gradient after an integration-by-parts convention is fixed. The proposition does not identify K_i^{div} with a Ricci tensor; it supplies a geometrically structured alternative to the direct local aggregation (8).

Remark (Choice of scales, conductances, and masses). *The inverse loop scales s_σ^{-1} determine the scale of individual curvature observations, whereas w_{ij} determines how edge fluxes are aggregated at a vertex. These roles should not be conflated. The formal construction requires only symmetric nonnegative conductances and positive masses; one may take $\mu_i = 1$ in the absence of a vertex-volume model. When the node metrics are represented in a shared reference frame, a possible conductance prior is*

$$w_{ij} = \exp(-\beta \text{d}_{\text{SPD}}(g_i, g_j)^2), \quad \beta > 0.$$

For a locally gauge-equivariant model without such a shared representation, w_{ij} should instead be supplied by gauge-invariant edge data or by a separately designed equivariant module. Keeping

these distinctions explicit prevents a coordinate-dependent prior from being mistaken for intrinsic graph geometry.

4. RICCI-TYPE METRIC RESPONSES AND EVOLUTION

4.1. Symmetric rectification of loop curvature. The lower-index matrix $g_i K_i$ is antisymmetric by (5). Hence neither $g_i K_i$ nor its direct sum can be a velocity of a symmetric metric. This is the obstruction that separates a loop-curvature observation from a Ricci-type tensor. The following elementary construction resolves the symmetry mismatch.

Definition 4.1 (Commutator Ricci-type response). *Let $B_{\mathcal{C},i} \in \text{Sym}_n$ be a symmetric response matrix assigned to a loop $\mathcal{C} \in \mathcal{C}_i$. Define*

$$(22) \quad \text{Ric}_g^{\text{D}}(i) := \sum_{\mathcal{C} \in \mathcal{C}_i} \alpha_{\mathcal{C}} (B_{\mathcal{C},i} g_i \Omega_{\mathcal{C}} - g_i \Omega_{\mathcal{C}} B_{\mathcal{C},i}).$$

If a single response matrix $B_i \in \text{Sym}_n$ is used at node i , this becomes

$$(23) \quad \text{Ric}_g^{\text{D}}(i) = [B_i, g_i K_i].$$

The divergence-type version is

$$(24) \quad \text{Ric}_g^{\text{D},\text{div}}(i) := [B_i, g_i K_i^{\text{div}}],$$

where K_i^{div} is the covariant divergence of the curvature edge cochain defined in (17).

Proposition 3.1 explains the geometric role of local loop aggregation. In the smooth calibration regime, a normalized logarithmic holonomy records curvature sampled in two local directions, while aggregation over the local loop family combines the available directional observations. When the discrete directions satisfy a frame-resolution condition, this mechanism approximates the Ricci contraction. On a general abstract graph, however, such a condition is not canonical; hence K_i is regarded as a local curvature aggregate rather than a discrete Ricci tensor.

Proposition 4.1 (Symmetry and gauge equivariance). *For every i , the matrices $\text{Ric}_g^{\text{D}}(i)$ and $\text{Ric}_g^{\text{D},\text{div}}(i)$ are symmetric. If the data are transformed by (7) and the response matrices obey $B'_{\mathcal{C},i} = Q_i B_{\mathcal{C},i} Q_i^{\text{T}}$, then*

$$(25) \quad \text{Ric}_{g'}^{\text{D}}(i) = Q_i \text{Ric}_g^{\text{D}}(i) Q_i^{\text{T}}, \quad \text{Ric}_{g'}^{\text{D},\text{div}}(i) = Q_i \text{Ric}_g^{\text{D},\text{div}}(i) Q_i^{\text{T}}.$$

Proof. For symmetric B and g_i -skew-adjoint Ω , the matrix $g_i \Omega$ is antisymmetric. Consequently

$$[B, g_i \Omega]^{\text{T}} = (B g_i \Omega - g_i \Omega B)^{\text{T}} = B g_i \Omega - g_i \Omega B.$$

Summation proves symmetry. Under (7), both B and $g_i \Omega$ transform by $Q_i(\cdot) Q_i^{\text{T}}$; the commutator and each weighted sum therefore transform as in (25). $\square$

The most basic nonlearnable choice is $B_{\mathcal{C},i} = g_i$. It gives

$$(26) \quad \text{Ric}_g^{\text{D}}(i) = \sum_{\mathcal{C} \in \mathcal{C}_i} \alpha_{\mathcal{C}} [g_i, g_i \Omega_{\mathcal{C}}].$$

More flexibly, $B_{\mathcal{C},i}$ can be a learned symmetric matrix. This parameter does not create curvature: curvature remains encoded in loop holonomy. Rather, $B_{\mathcal{C},i}$ specifies how the curvature observation couples to the metric degrees of freedom. The formula can vanish for commuting $B_{\mathcal{C},i}$ and $g_i \Omega_{\mathcal{C}}$, which is an expected feature of a commutator-based response, not an error to be hidden.

4.2. SPD-preserving metric evolution. Since the tangent space at an SPD matrix consists of symmetric matrices, Proposition 4.1 makes the response eligible to drive a metric update. A Euclidean update $g_i^+ = g_i - 2\eta \text{Ric}_g^D(i)$ may nevertheless leave the SPD cone. We therefore use the exponential retraction

$$(27) \quad g_i^+ = g_i^{1/2} \exp\left(-2\eta g_i^{-1/2} \text{Ric}_g^D(i) g_i^{-1/2}\right) g_i^{1/2}, \quad \eta > 0.$$

The matrix in the exponential is symmetric, so (27) is SPD for every step size. Replacing Ric_g^D by $\text{Ric}_g^{D,\text{div}}$ gives the divergence-type evolution. This is a discrete Ricci-type flow: it shares the form of a curvature-driven metric evolution with the smooth Ricci flow equation $\partial_t g_t = -2 \text{Ric}(g_t)$, but its generator depends on the selected graph loops, their normalizations, and the response matrices.

Algorithm 1 Holonomy-based Ricci-type metric update at one layer

Input: Graph G , metrics $\{g_i\}$, edge transports $\{F_{ij}\}$, oriented face set $\mathcal{F}$ (one orientation per selected underlying triangle) with loop collection $\{\mathcal{C}_i\}$, symmetric response matrices $\{B_{\mathcal{C},i}\}$, and step size η .

Initialize: Updated SPD metrics $\{g_i^+\}$.

- 1: **for** each oriented triangle $\mathcal{C} = (i, j, k, i)$ **do**
 - 2: Compute $H_{\mathcal{C}} = F_{ik}F_{kj}F_{ji}$ and $\Omega_{\mathcal{C}} = -s_{\mathcal{C}}^{-1} \text{Log}(H_{\mathcal{C}})$.
 - 3: **end for**
 - 4: **for** each node i **do**
 - 5: Form $S_i \leftarrow \sum_{\mathcal{C} \in \mathcal{C}_i} \alpha_{\mathcal{C}} [B_{\mathcal{C},i}, g_i \Omega_{\mathcal{C}}]$.
 - 6: Set $g_i^+ \leftarrow g_i^{1/2} \exp(-2\eta g_i^{-1/2} S_i g_i^{-1/2}) g_i^{1/2}$.
 - 7: **end for**
 - 8: **Output:** $\{g_i^+\}$.
-

5. A GEOMETRY-AWARE LEARNING REALIZATION

The preceding sections define a family of geometric layers rather than a fully specified neural architecture. This section records one conservative way to make the free choices learnable while retaining the structural constraints. Let $h_i^{(\ell)}$ denote semantic node features and $g_i^{(\ell)} \in \text{SPD}_n$ metric features at layer ℓ . A model may learn orthogonal factors $O_{ij}^{(\ell)}$ and set

$$(28) \quad F_{ij}^{(\ell)} = (g_i^{(\ell)})^{-1/2} O_{ij}^{(\ell)} (g_j^{(\ell)})^{1/2}, \quad O_{ij}^{(\ell)} \in \text{O}(n), \quad O_{ji}^{(\ell)} = (O_{ij}^{(\ell)})^\top,$$

so compatibility is exact at every layer. In practice, an orthogonal factor can be produced by an exponential of a skew-symmetric matrix or by an orthogonalization parameterization.

Likewise, a response matrix can be parameterized as

$$(29) \quad B_{\mathcal{C},i}^{(\ell)} = L_{\mathcal{C},i}^{(\ell)} (L_{\mathcal{C},i}^{(\ell)})^\top + \delta I, \quad \delta > 0,$$

which guarantees both symmetry and positive definiteness. The curvature response is then inserted into (27). The semantic features can be updated by an ordinary equivariant or invariant message-passing operator that is conditioned on scalar loop statistics, for example norms of $\Omega_{\mathcal{C}}^{(\ell)}$ or gauge-invariant contractions constructed from them.

If a common registration of node metric coordinates is part of the data model, one possible symmetric conductance prior and its associated row-normalized attention are

$$(30) \quad \tilde{w}_{ij}^{(\ell)} = \exp\left[-\beta_\ell \text{dSPD}(g_i^{(\ell)}, g_j^{(\ell)})^2\right], \quad \pi_{ij}^{(\ell)} = \frac{\tilde{w}_{ij}^{(\ell)}}{\sum_{k \sim i} \tilde{w}_{ik}^{(\ell)}}.$$

Here $\tilde{w}_{ij}^{(\ell)} = \tilde{w}_{ji}^{(\ell)}$ may be used as the conductance w_{ij} in the divergence construction, whereas $\pi_{ij}^{(\ell)}$ is a directional attention weight for semantic aggregation. For unregistered fibres, Equation (30) should not be used as a gauge-invariant rule. This is not merely a technical qualification:

the separation between a common-coordinate metric prior and local-frame gauge symmetry determines what can legitimately be claimed about the model.

The full layer can therefore be read in two stages. First, edge transports produce loop holonomies and a symmetric metric response, which updates $g_i^{(\ell)}$ within SPD_n . Second, node semantics $h_i^{(\ell)}$ are propagated using a task-specific operator conditioned on the resulting geometry. This division respects the different roles of metric evolution and semantic aggregation. It also keeps a learning system from conflating an arbitrary edge transformation with a metric-compatible parallel transport.

6. STRUCTURAL CHECKS AND SPECIAL CASES

This section records several elementary checks that determine how the proposed objects should behave. They are not convergence results, but they distinguish the holonomy construction from an arbitrary matrix-valued message-passing rule.

Flat-loop consistency. Suppose that $H_{\mathcal{C}} = I$ for every selected loop based at i . The principal logarithm of I is zero. Therefore every loop curvature term $\Omega_{\mathcal{C}}$ vanishes, and so do the sums in (8) and (17). The commutator definitions (22) and (24) then vanish. In (27), the exponential has zero exponent. Thus

$$K_i = 0, \quad K_i^{\text{div}} = 0, \quad \text{Ric}_g^{\text{D}}(i) = 0, \quad \text{Ric}_g^{\text{D,div}}(i) = 0.$$

Consequently, both updates defined by (27) leave g_i unchanged.

Existence of connection data. For arbitrary SPD metrics, metric-compatible transports always exist: choosing $O_{ij} = I$ in (2) supplies one. This choice is globally flat, however, because the transports telescope around every loop. Nontrivial curvature is therefore carried by the incompatible edgewise orthogonal factors O_{ij} , not by the node metrics alone. This observation has a useful modelling consequence. An architecture that learns only $\{g_i\}$ while fixes $O_{ij} = I$ cannot produce nonzero holonomy curvature; it needs either learned connection degrees of freedom or a transport rule derived from external geometric data.

At the opposite extreme, a graph with no selected cycles has $\mathcal{C}_i = \emptyset$ at every node. Its direct and divergence-type curvature responses are zero by definition. This is mathematically consistent but restrictive for tree-like graphs. There are at least three principled extensions: use a supplied two-dimensional cell complex; construct a local clique or Vietoris–Rips completion whose added faces are part of the model; or use longer fundamental cycles with an explicitly chosen scale normalization. None of these extensions is silently assumed in the present triangular construction.

Role of the response matrix. The response matrix B must be distinguished from the transport connection. The transport determines $H_{\mathcal{C}}$ and hence the curvature observation, while B selects a symmetric channel through which that observation changes the metric. Its nonuniqueness is unavoidable in the current graph setting because a graph alone does not provide a canonical analogue of the smooth tensor contraction defining the Ricci tensor. Two illustrative cases are worth separating:

- *Fixed response.* Taking $B_{\mathcal{C},i} = g_i$ yields the completely specified symmetric response in (26). This is a useful baseline when one wants the fewest learned geometric parameters.
- *Learned response.* Taking $B_{\mathcal{C},i}$ from (29) permits loop- and node-dependent coupling while preserving symmetry and gauge equivariance. It enlarges the model class and should be evaluated against the fixed baseline rather than treated as a theorem about a unique curvature contraction.

In either case, the commutator makes clear that the metric response detects noncommutativity between the curvature channel $g_i\Omega_{\mathcal{C}}$ and the selected symmetric channel $B_{\mathcal{C},i}$. It is this spectral mismatch that drives the proposed evolution.

7. GEOMETRIC CALIBRATION AND SYNTHETIC CONSISTENCY CHECKS

This section tests the finite-dimensional mechanism of the construction on a geometry with known curvature. The tests have deliberately separated roles. The first calibrates the passage

from loop holonomy to curvature on the unit sphere; the second verifies that the full

$$g_i, O_{ij} \mapsto F_{ij} = g_i^{-1/2} O_{ij} g_j^{1/2} \mapsto H_C \mapsto \Omega_C$$

representation preserves that curvature after comparison in a common orthonormal frame; and the third gives a synthetic, non-oracle realization of the learnable factors in (28). No task labels are used in these checks. Their purpose is to make the geometric mechanism falsifiable, rather than to substitute for a continuum theorem or an application-level performance evaluation.

Throughout this section, the graph is an oriented triangular icosphere approximating the unit sphere $\mathbb{S}^2$. The fibre at p_i is $T_{p_i}\mathbb{S}^2$, represented in a local orthonormal frame. The controlled geometry supplies the unsigned spherical area $A_C > 0$ of every face, and we make the particular choice $s_C = A_C$. This is a known-geometry specialization of the effective scale in Definition 2.3; it is not a prescription for the task-dependent normalization of a general graph.

7.1. Holonomy–curvature calibration on the sphere. We first take $g_i = I$ and let O_{ij} be the exact Levi–Civita parallel transport along the shortest great-circle arc from p_j to p_i . Thus $F_{ij} = O_{ij}$. For an outward-oriented spherical triangle, let θ_C denote the signed principal rotation angle of H_C , with the sign chosen consistently with the face orientation. On the unit sphere, the exact relation is

$$\theta_C = A_C, \quad \hat{K}_C := \frac{\theta_C}{A_C} = 1.$$

This scalar presentation is compatible with the matrix convention $\Omega_C = -A_C^{-1} \text{Log}(H_C)$: the latter retains the oriented endomorphism, whereas $\hat{K}_C$ records its scalar holonomy–area calibration.

Figure 1 reports the calculation at four mesh levels, from 20 to 1280 faces. The largest observed $|\theta_C - A_C|$ is below 4×10^{-16} , and the estimated scalar curvature is 1 to numerical precision. The deliberately incorrect flat control $F_{ij} = I$ has identically zero loop holonomy, so it cannot recover the curvature of the sphere. The refinement panel is included only as a numerical stability check of the exact construction: since the transports and areas are analytically supplied, it is not evidence for a general convergence rate under mesh refinement.

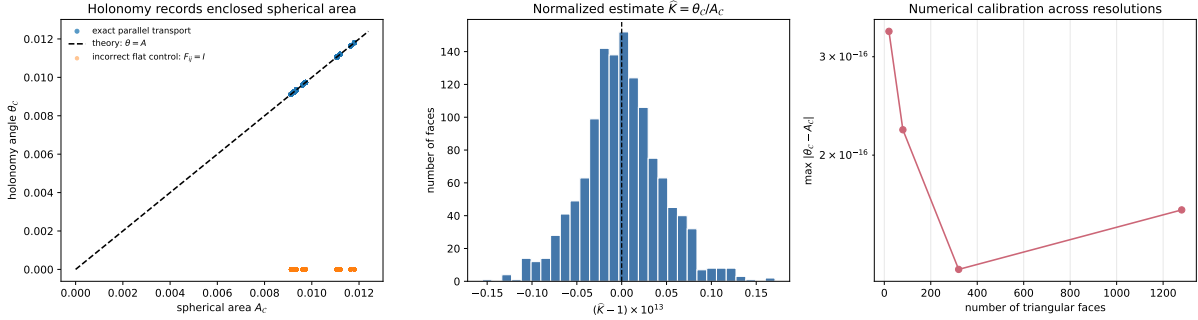


FIGURE 1. Holonomy–curvature calibration on the unit sphere. Exact Levi–Civita edge transport gives the signed relation $\theta_C = A_C$ for every oriented spherical face. With the controlled scale $s_C = A_C$, the scalar estimate $\hat{K}_C = \theta_C / A_C$ equals the unit-sphere curvature to numerical precision. The flat control $F_{ij} = I$ produces zero holonomy.

7.2. Metric representation calibration. The preceding test isolates the holonomy–curvature relation but uses the trivial fibre metric. We next use a smooth, spatially varying SPD field $\{g_i\}$ with condition numbers between 1.003 and 2.600 on the finest mesh, while retaining the same exact sphere transports $\{O_{ij}\}$. We then form F_{ij} by (2). This is the full metric-compatible representation used in the paper, but it is not asserted to be the Levi–Civita connection of the auxiliary field $\{g_i\}$.

For a loop based at i , the metric factors telescope exactly:

$$H_C^F = g_i^{-1/2} H_C^O g_i^{1/2}, \quad \tilde{\Omega}_C^F := g_i^{1/2} \Omega_C^F g_i^{-1/2} = \Omega_C^O.$$

The first equality makes clear why a raw matrix comparison is not geometric: Ω_C^F and Ω_C^O are written in different metric representations. On the finest mesh, the unwhitened difference $\|\Omega_C^F - \Omega_C^O\|_F$ ranges from 2.09×10^{-3} to 7.21×10^{-1} , whereas the whitened difference $\|\tilde{\Omega}_C^F - \Omega_C^O\|_F$ is at most 3.5×10^{-13} ; see Figure 2. The compatibility, telescoping, and g_i -skew residuals are also at numerical precision. Thus the nontrivial local metrics do not introduce an additional holonomy distortion when O_{ij} and s_C are fixed.

This conclusion is intentionally limited. It does not say that g_i is irrelevant in the full model: the local metric may still enter the loop scales, weights, response matrices, and SPD evolution. It only isolates the representation identity that underlies the construction (2).

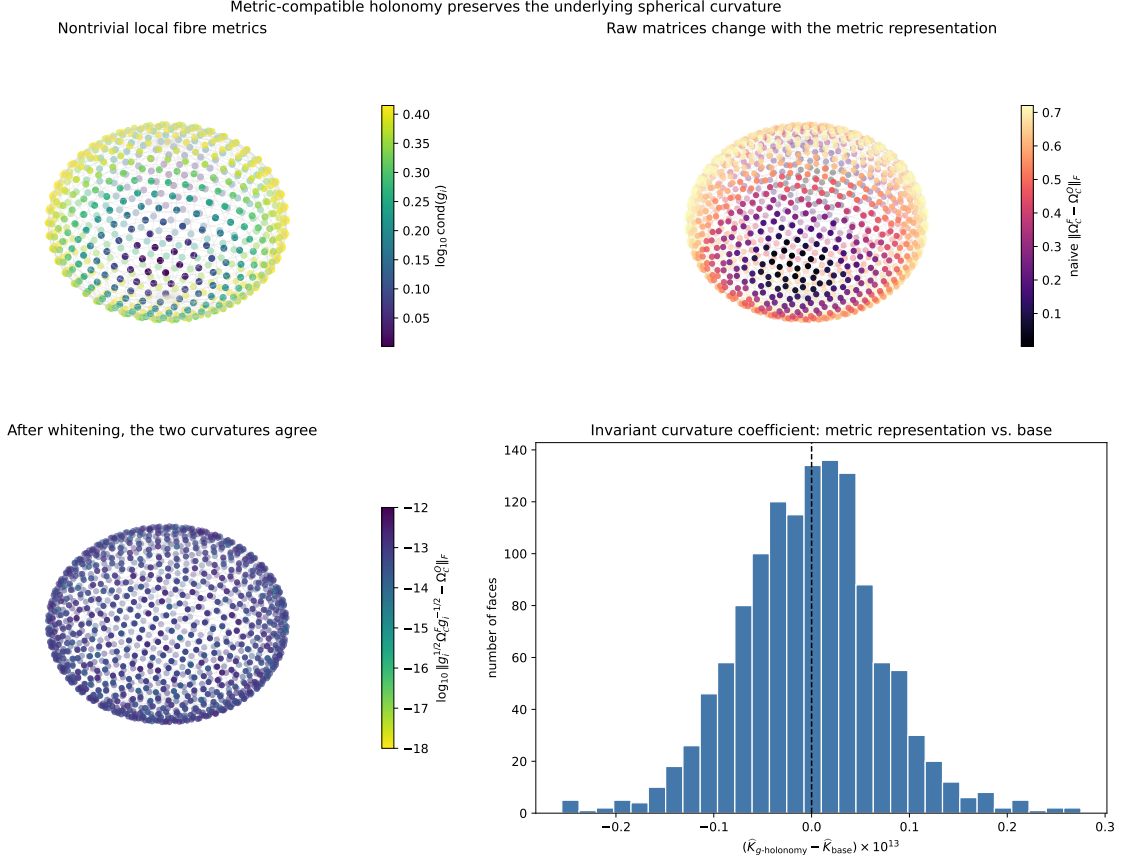


FIGURE 2. Metric-compatible representation calibration. A nontrivial local SPD field changes the raw matrix representation of loop curvature, but the whitened curvature $g_i^{1/2} \Omega_C^F g_i^{-1/2}$ agrees with the orthonormal-frame sphere curvature Ω_C^O to numerical precision.

7.3. Synthetic recovery of learned edge transports. The previous two checks use the exact sphere transport and hence do not test the learnable degree of freedom in Section 5. We therefore construct a small synthetic recovery problem on a fixed level-two icosphere (162 vertices, 480 undirected edges, and 320 oriented faces). For each directed edge $i \rightarrow j$, the estimator observes M noisy local vector correspondences

$$x_{ij}^{(m)} \in \mathbb{R}^2, \quad y_{ji}^{(m)} = O_{ji}^* x_{ij}^{(m)} + \eta_{ij}^{(m)}, \quad m = 1, \dots, M,$$

where O_{ji}^* is the hidden exact sphere transport and the noise standard deviation is 0.15. The hidden transport is used only to generate the synthetic observations and to evaluate the result. It is not an input to the estimator.

For each edge, we use the orthogonal Procrustes rule

$$\hat{O}_{ji} = \arg \min_{O \in \text{SO}(2)} \sum_{m=1}^M \|Ox_{ij}^{(m)} - y_{ji}^{(m)}\|_2^2, \quad \hat{O}_{ij} = \hat{O}_{ji}^\top,$$

followed by the metric-compatible parameterization $\hat{F}_{ij} = g_i^{-1/2} \hat{O}_{ij} g_j^{1/2}$. This is one explicit, CPU-scale realization of the orthogonal-factor learning option in (28); it is not claimed to be the unique learning architecture of the paper. We use the same nontrivial metric field as above, set $s_C = A_C$, and average over eight independent noise realizations.

TABLE 1. Synthetic learned-transport calibration on the fixed sphere mesh. Each entry is the mean over eight noise realizations. The curvature error is measured after whitening at the loop base point: $\|\tilde{\Omega}_C^F - \Omega_C^*\|_F$.

Correspondences per edge M	1	4	16	64	256
Mean transport error $\ \hat{O} - O^*\ _F$	2.094×10^{-1}	6.683×10^{-2}	3.123×10^{-2}	1.524×10^{-2}	7.488×10^{-3}
Mean whitened curvature error	1.125×10^1	3.024	1.404	6.883×10^{-1}	3.437×10^{-1}

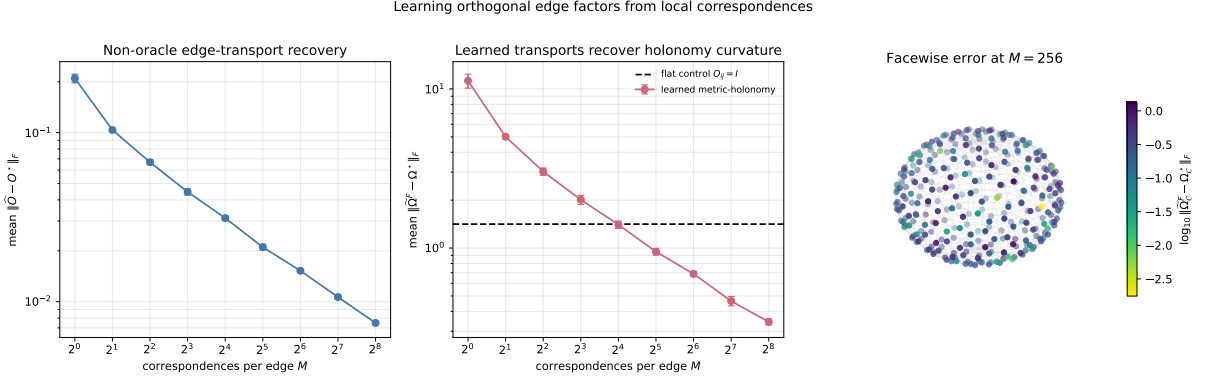


FIGURE 3. Synthetic recovery of non-oracle edge transports. Per-edge $\text{SO}(2)$ Procrustes estimates use only noisy vector correspondences. As the number M of correspondences increases, both the learned transport error and the induced whitened holonomy-curvature error decrease. The dashed line is the flat connection control $O_{ij} = I$.

Table 1 and Figure 3 show that, under this observation model, both the edge transport error and the induced whitened curvature error decline as M increases. At $M = 256$, the mean curvature error is 3.44×10^{-1} , compared with 1.41 for the flat control $O_{ij} = I$. The metric-representation residual remains at numerical precision throughout, so the observed error is attributable to imperfect transport recovery rather than to the metric-compatible parameterization.

This is an empirical fixed-mesh consistency trend as the number of correspondences increases. It does not establish a limit as the mesh size tends to zero, nor does it show that an analogous training signal is available in an arbitrary application. In particular, a mesh-refinement theorem would require explicit control of the learned edge error relative to the face scale.

Scope of the synthetic checks. The exact algebraic properties proved in the preceding sections—metric compatibility, reversibility, telescoping of the loop holonomy, and local orthogonal gauge equivariance—are not inferred from the numerical plots. They hold by construction under their stated assumptions. The present checks instead verify an implementation on a geometry with known curvature and exhibit one concrete route by which observational information can determine the otherwise free orthogonal factors. A subsequent task-level experiment must separately test whether the resulting geometric information is useful in data analysis.

8. DISCUSSION, SCOPE, AND NEXT STEPS

The construction proposed here turns a basic fact of connection geometry into a graph modelling device: nontrivial loop holonomy supplies a local curvature observation. Metric compatibility locates this observation in a g_i -orthogonal Lie algebra, and the commutator response converts its lowered-index antisymmetry into a symmetric metric direction. This produces a well-defined SPD-preserving update and makes the role of every additional choice visible.

Several limitations delimit the current result. First, triangles are used as minimal loops, but many graphs have few or no triangles. Longer cycles or a chosen cell-complex completion would be needed in sparse settings; their scale normalization cannot be inferred from the present construction alone. Second, the logarithm requires a branch choice and can become numerically unstable near the excluded spectrum. Third, the response matrix B is a modelling choice, not a replacement for the smooth Ricci contraction. Finally, the calibrations in Section 7 do not provide a general continuum limit, a stability theorem for learned dynamics, or a predictive advantage on a learning task. These questions require separate hypotheses and validation.

There are several concrete directions for completing the theory and evaluating the model. A cell-complex version could provide an explicit discrete area and orientation calculus. A typed tensor formulation may extend the present orthogonal gauge covariance to broader changes of local frame. On the learning side, one should compare fixed versus learned response matrices, test the effect of loop scales, and evaluate whether curvature-conditioned diffusion improves task performance without destabilizing metric evolution. These are natural continuations of the graph-geometric program, but they are not assumed in the statements of this paper.

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